

Citations of Retracted ICPSR Papers

Number of citations:

```
# number of citations per paper
refs |> group_by(filepath) |> count()
```

```
## # A tibble: 9 x 2
## # Groups:   filepath [9]
##   filepath      n
##   <chr>      <int>
## 1 ECigUse      61
## 2 HowDidTheyDecide    4
## 3 HypertensionGuidelines 50
## 4 MarijuanaUse      80
## 5 NonmetroRole      11
## 6 SchoolSocialBonds 267
## 7 SuicideNatDisasters 178
## 8 TurningIn        16
## 9 WorldsApart     465
```

```
# add retraction years to refs
refs$ret.year <- numeric(length = nrow(refs))
refs$ret.year[which(refs$filepath == "ECigUse")] <- merged$RetractionYear[5]
refs$ret.year[which(refs$filepath == "HowDidTheyDecide")] <- merged$RetractionYear[6]
refs$ret.year[which(refs$filepath == "HypertensionGuidelines")] <- merged$RetractionYear[9]
refs$ret.year[which(refs$filepath == "MarijuanaUse")] <- merged$RetractionYear[7]
refs$ret.year[which(refs$filepath == "NonmetroRole")] <- merged$RetractionYear[2]
refs$ret.year[which(refs$filepath == "SchoolSocialBonds")] <- merged$RetractionYear[3]
refs$ret.year[which(refs$filepath == "SuicideNatDisasters")] <- merged$RetractionYear[4]
refs$ret.year[which(refs$filepath == "TurningIn")] <- merged$RetractionYear[8]
refs$ret.year[which(refs$filepath == "WorldsApart")] <- merged$RetractionYear[1]
```

```
# number of citations per paper by year
refsbyyear <- refs |> group_by(filepath, year, ret.year) |>
  summarise(num_citations = n())
```

```
## `summarise()` has grouped output by 'filepath', 'year'. You can override using
## the `.groups` argument.
```

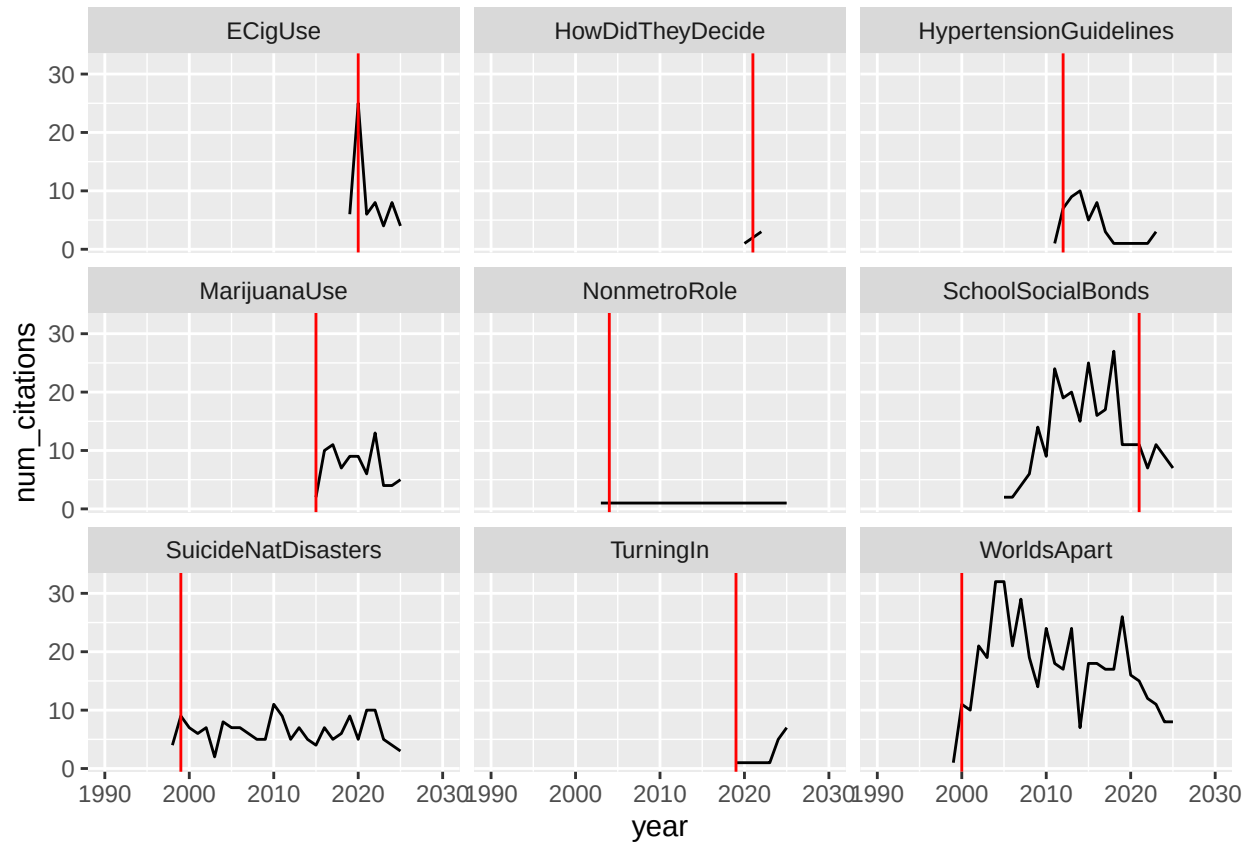
```
(head(refsbyyear, 20))
```

```
## # A tibble: 20 x 4
## # Groups:   filepath, year [20]
##   filepath      year ret.year num_citations
##   <chr>      <dbl>   <dbl>         <int>
## 1 ECigUse    2019     2020           6
## 2 ECigUse    2020     2020          25
## 3 ECigUse    2021     2020           6
## 4 ECigUse    2022     2020           8
## 5 ECigUse    2023     2020           4
```

##	6	ECigUse	2024	2020	8
##	7	ECigUse	2025	2020	4
##	8	HowDidTheyDecide	2020	2021	1
##	9	HowDidTheyDecide	2022	2021	3
##	10	HypertensionGuidelines	2011	2012	1
##	11	HypertensionGuidelines	2012	2012	7
##	12	HypertensionGuidelines	2013	2012	9
##	13	HypertensionGuidelines	2014	2012	10
##	14	HypertensionGuidelines	2015	2012	5
##	15	HypertensionGuidelines	2016	2012	8
##	16	HypertensionGuidelines	2017	2012	3
##	17	HypertensionGuidelines	2018	2012	1
##	18	HypertensionGuidelines	2019	2012	1
##	19	HypertensionGuidelines	2021	2012	1
##	20	HypertensionGuidelines	2022	2012	1

Plots of citations by year. Retraction Year in red.

```
# plot citations by year
refsbyyear |>
  group_by(filepath) |>
  ggplot(aes(x = year, y = num_citations)) +
    geom_line() +
    geom_vline(aes(xintercept = ret.year), colour = "red") +
    xlim(1990, 2030) +
    facet_wrap(~filepath)
```



Summary table - number of citations before and after retraction year:

```
refsbyyear |> group_by(filepath, year <= ret.year) |>
  count()
```

```
## # A tibble: 18 x 3
## # Groups:   filepath, year <= ret.year [18]
##   filepath      `year <= ret.year`      n
##   <chr>          <lgl>          <int>
## 1 ECigUse        FALSE            5
## 2 ECigUse        TRUE             2
## 3 HowDidTheyDecide FALSE            1
## 4 HowDidTheyDecide TRUE             1
## 5 HypertensionGuidelines FALSE          10
## 6 HypertensionGuidelines TRUE             2
## 7 MarijuanaUse   FALSE          10
## 8 MarijuanaUse   TRUE             1
## 9 NonmetroRole   FALSE            9
## 10 NonmetroRole   TRUE             2
## 11 SchoolSocialBonds FALSE            4
## 12 SchoolSocialBonds TRUE           17
## 13 SuicideNatDisasters FALSE          26
## 14 SuicideNatDisasters TRUE             2
## 15 TurningIn      FALSE            5
## 16 TurningIn      TRUE             1
## 17 WorldsApart    FALSE          25
## 18 WorldsApart    TRUE             2
```